Assignment Week 3B – Biological Database

Based on this week's lessons on biological databases, complete the following task and submit your answers by pasting them in the space provided.

Using the TCGA data portal:

1. **Search Criteria**
    1. **Disease**: Liver cancer
    2. **Primary site**: *Adenocarcinoma, NOS*
    3. **Experimental strategy**: *RNA-Seq*
2. **Using your search results, answer the following:**

    a. How many **mutated genes** are displayed?

    b. How many **total mutations** are present?

    c. Which gene has the **highest number of mutations**?

    d. What is the **mutation frequency** of this gene in liver cancer (in percentage)?

    e. What is the **amino acid change** and at what **position** did it occur? *(e.g., Alanine (A) → Arginine (R) at position 175)*

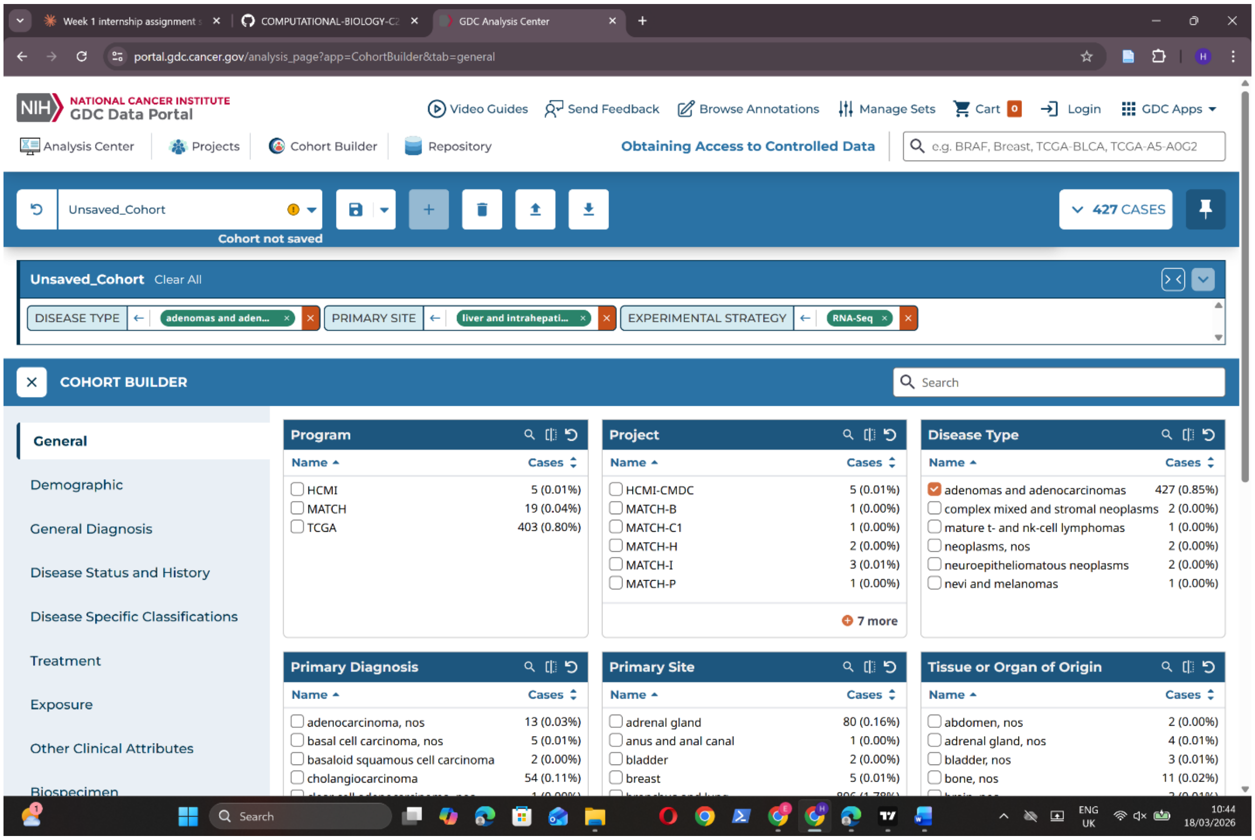
Biological Impact (Advanced Task - Optional)

1. Using **TCGA's 'Impact' column** (VEP, PolyPhen or SIFT):

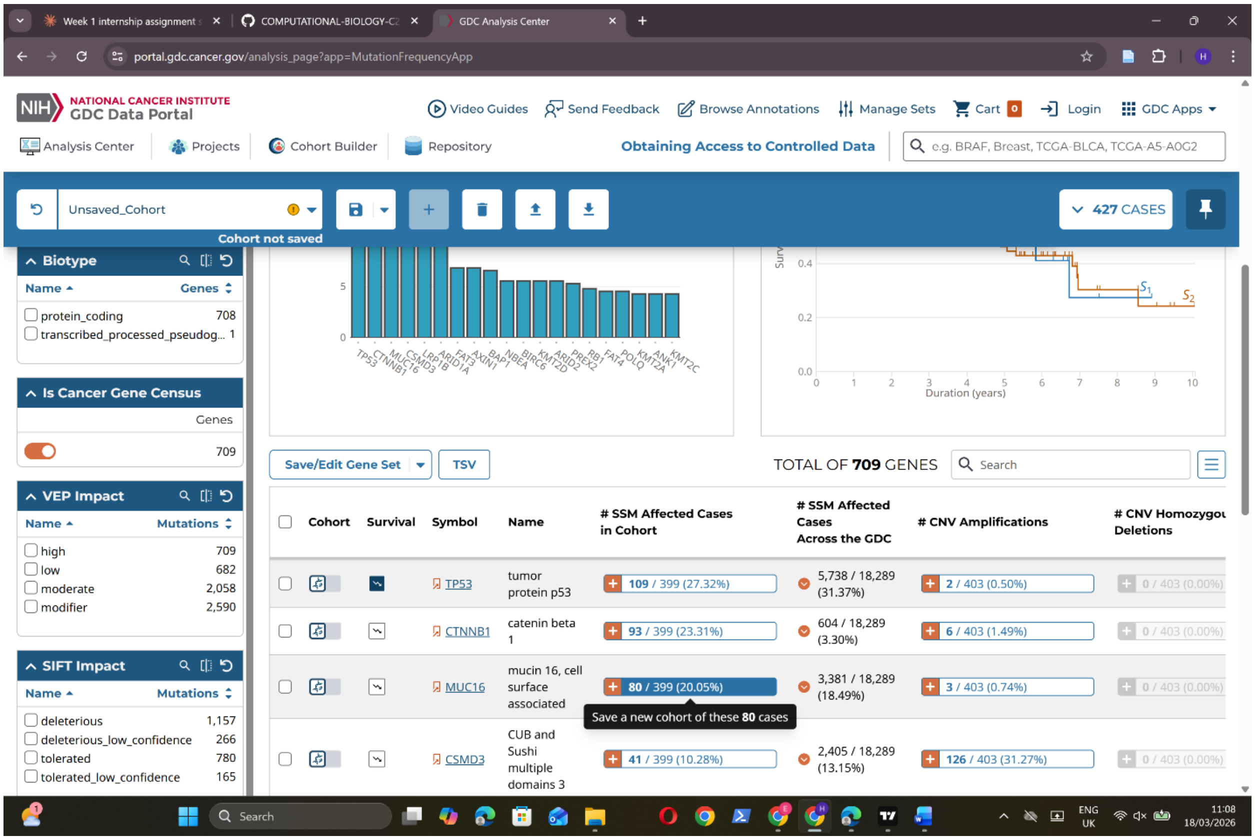
    f. What is the predicted effect of this mutation? *(*e.g., "probably damaging", "deleterious", "moderate impact"*)*

    g. How might this mutation affect the protein's structure or function? *(*Briefly explain in 2–3 sentences.*)*

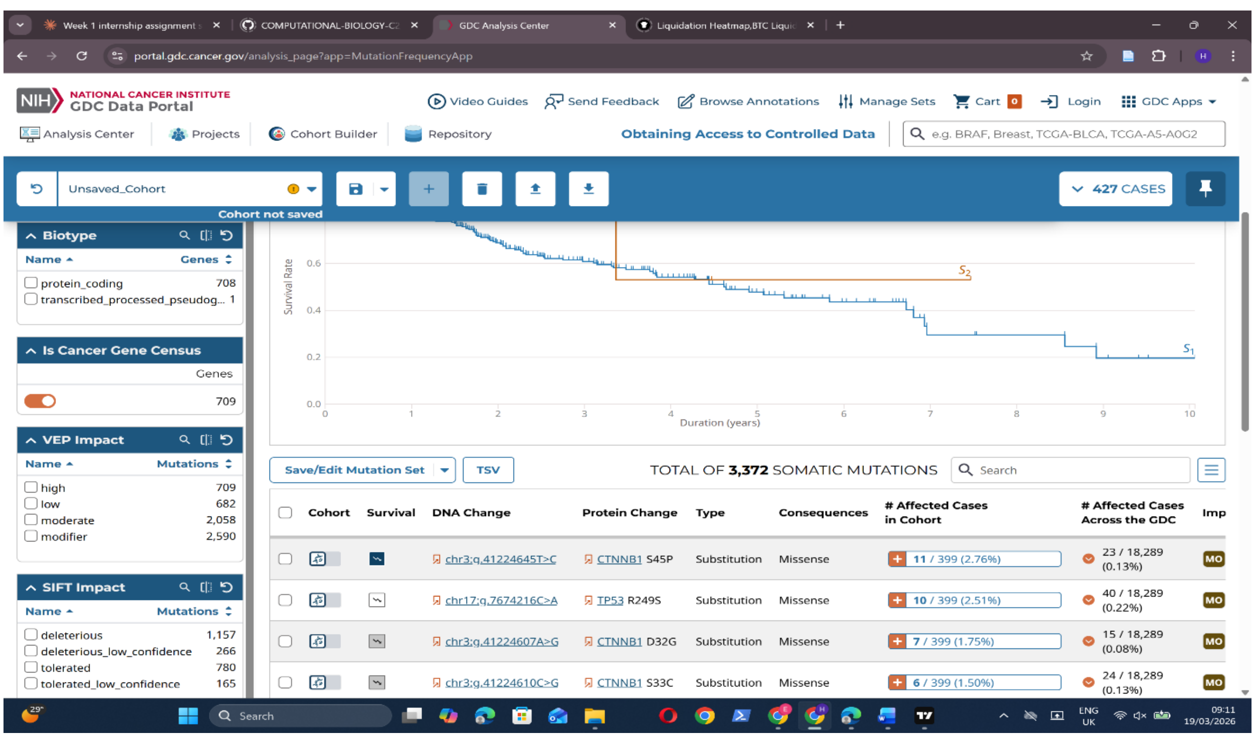
ANSWERS



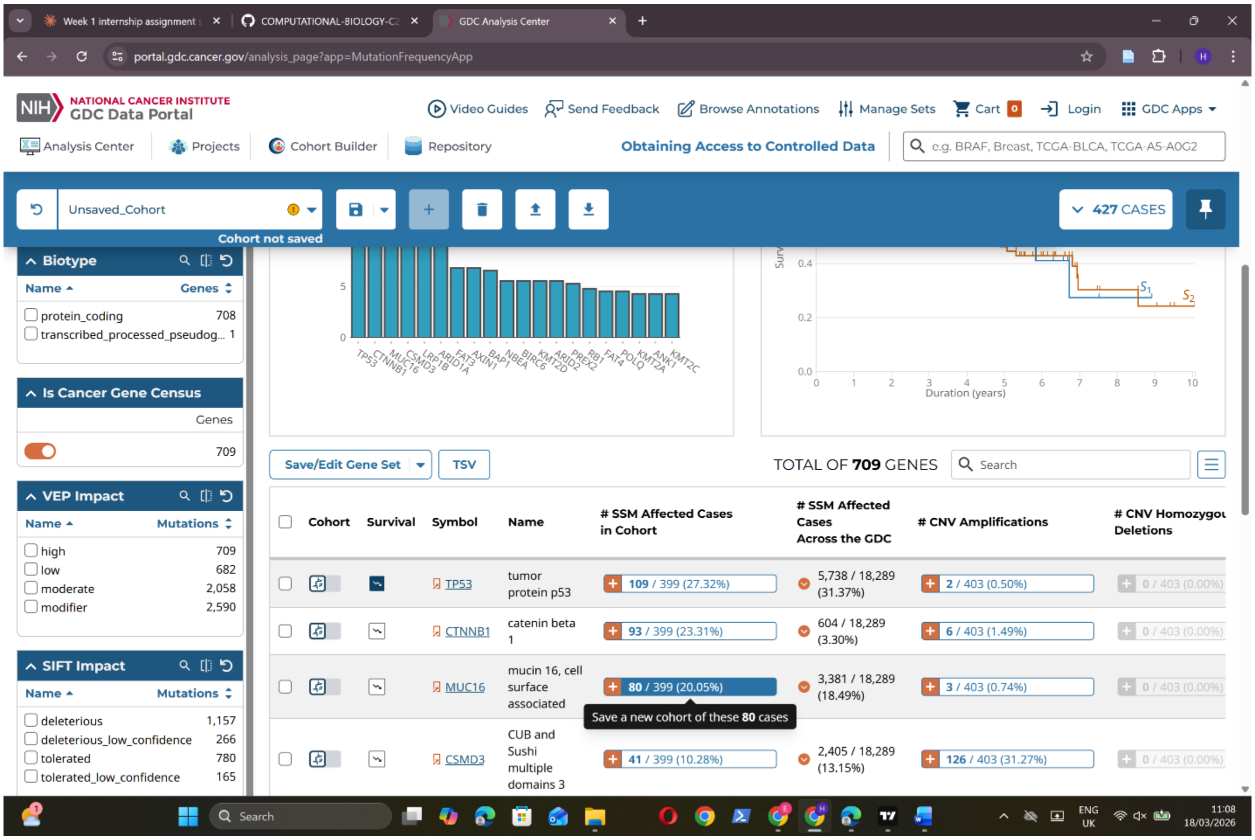
2.a. Mutated genes on display (709)



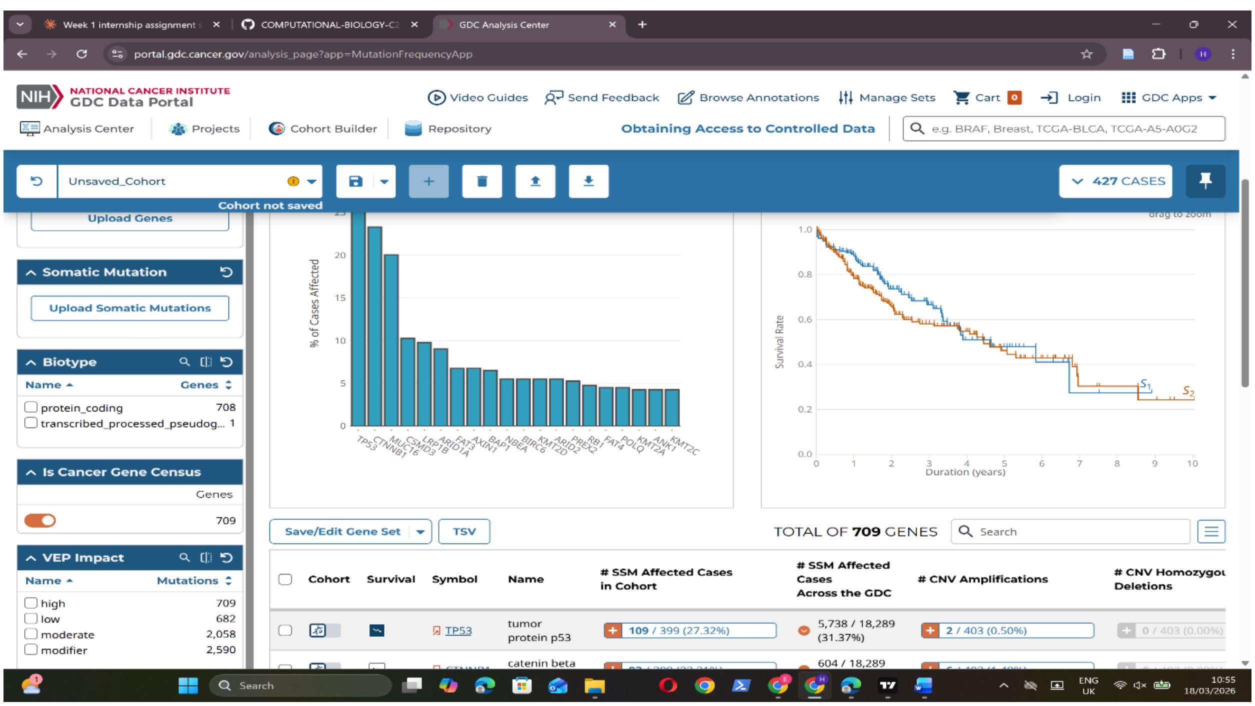
b. Total mutations (3,372)



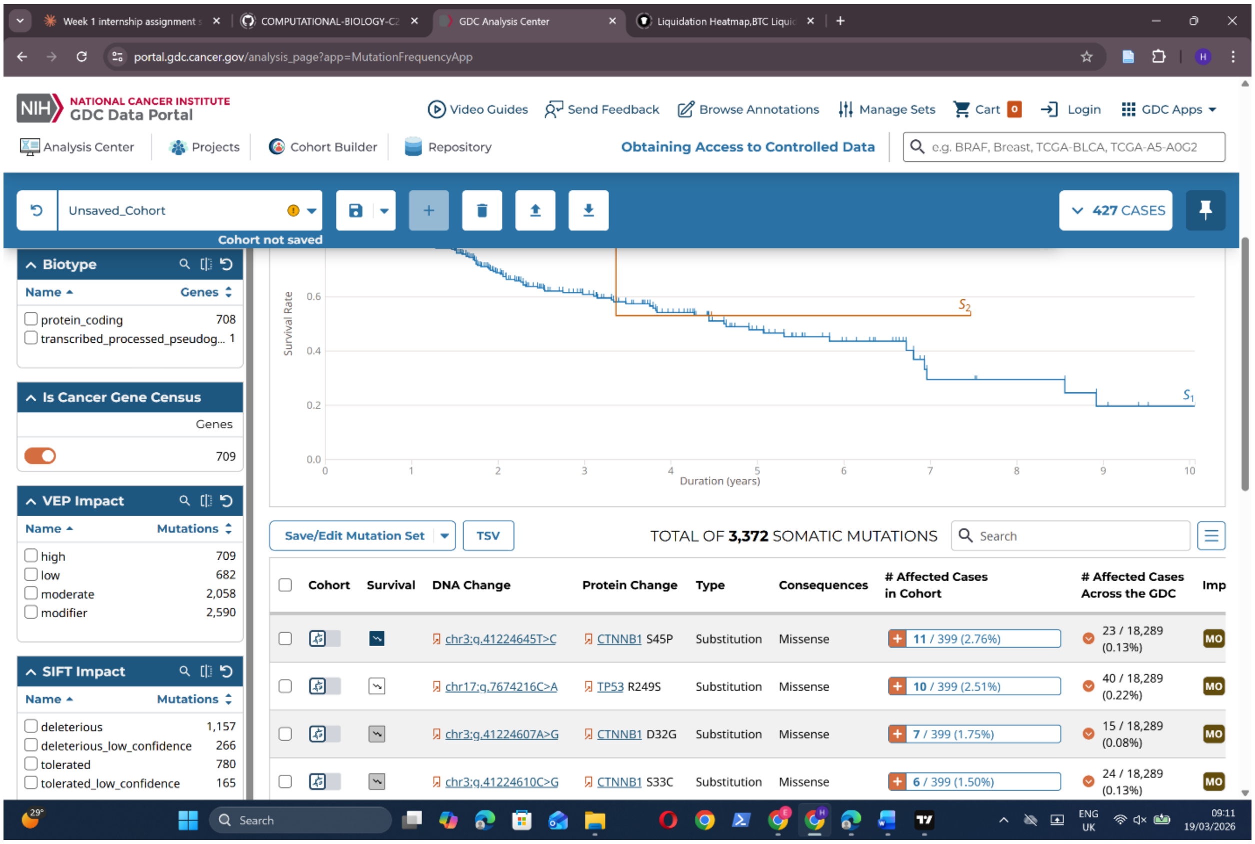
c. Gene with highest number of mutations (TP53 gene)



d. Mutation frequency (27.32%)

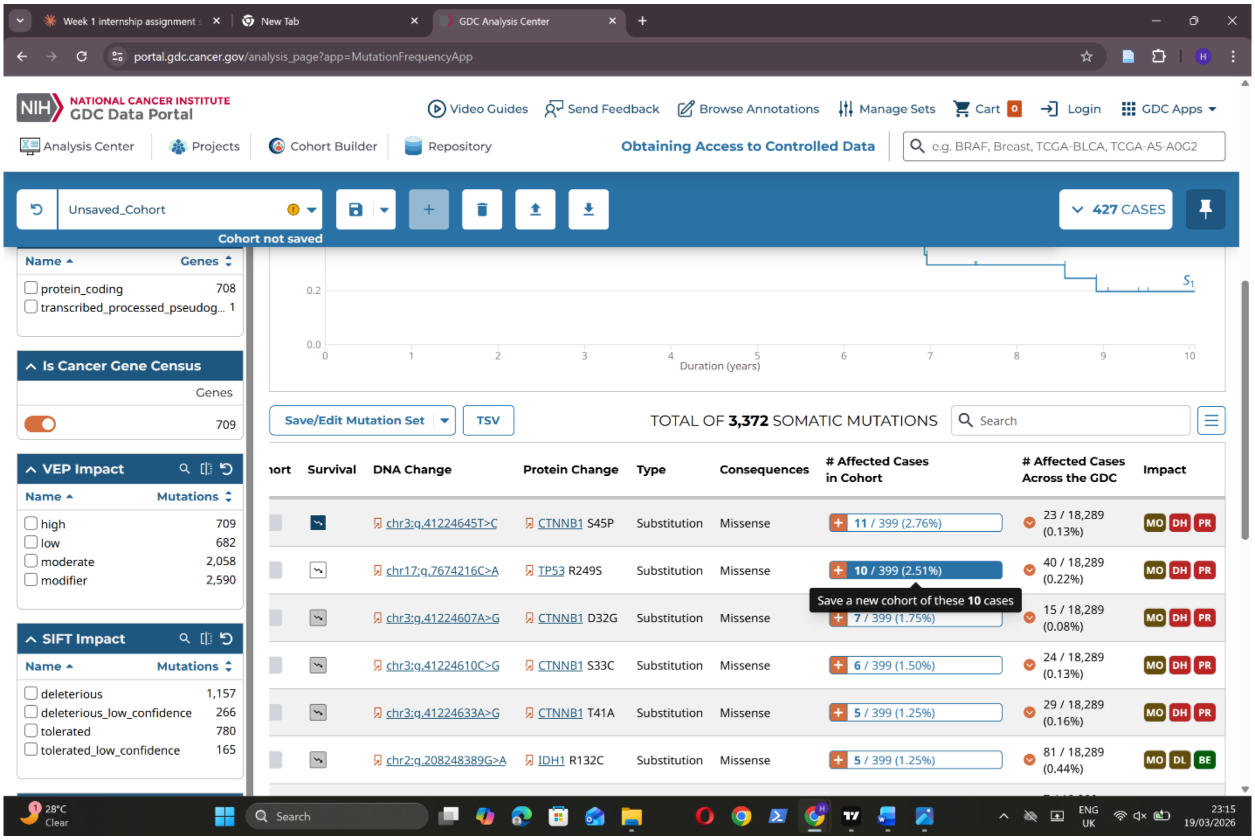


e. Amino acid change Arginine (R) → Serine (S) at position 249.



BIOLOGICAL IMPACT- ADVANCED TASK

2.f. VEP effect: Moderate



g. How this moderate effect affects the protein structure

TP53 is a tumor suppressor gene that codes for the p53 protein, often called the "guardian of the genome." In its normal state, p53 detects DNA damage and either pauses cell division for repair or triggers apoptosis. The R249S mutation disrupts p53's DNA-binding ability, rendering it unable to perform this function and allowing damaged cells to proliferate unchecked, contributing to liver cancer development.

References

Balasundaram, A., & Doss, C. G. P. (2022). Unraveling the Structural Changes in the DNA-Binding Region of Tumor Protein p53 (TP53) upon Hotspot Mutation p53 Arg248 by Comparative Computational Approach. *International Journal of Molecular Sciences*, *23*(24), 15499. https://doi.org/10.3390/ijms232415499